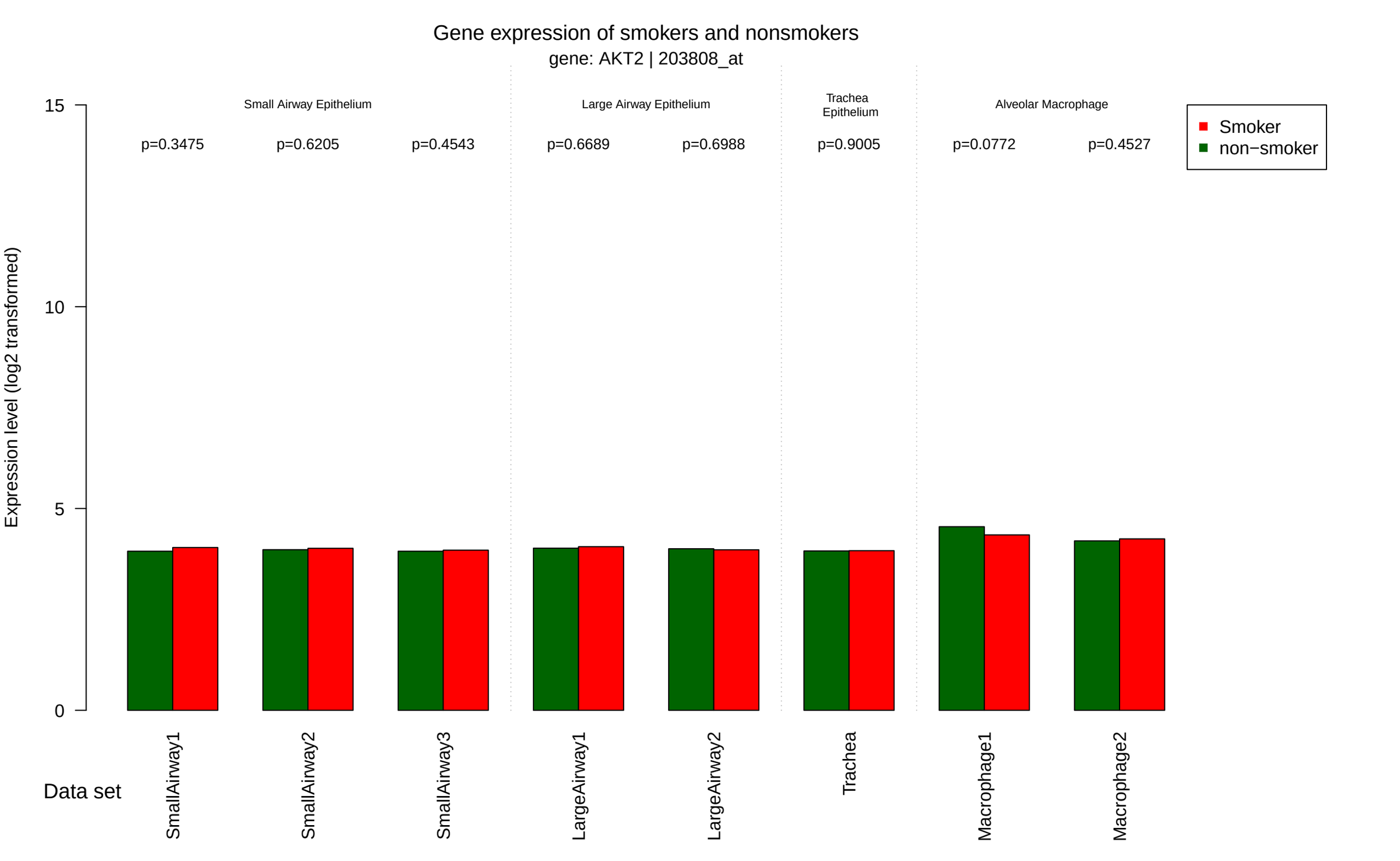
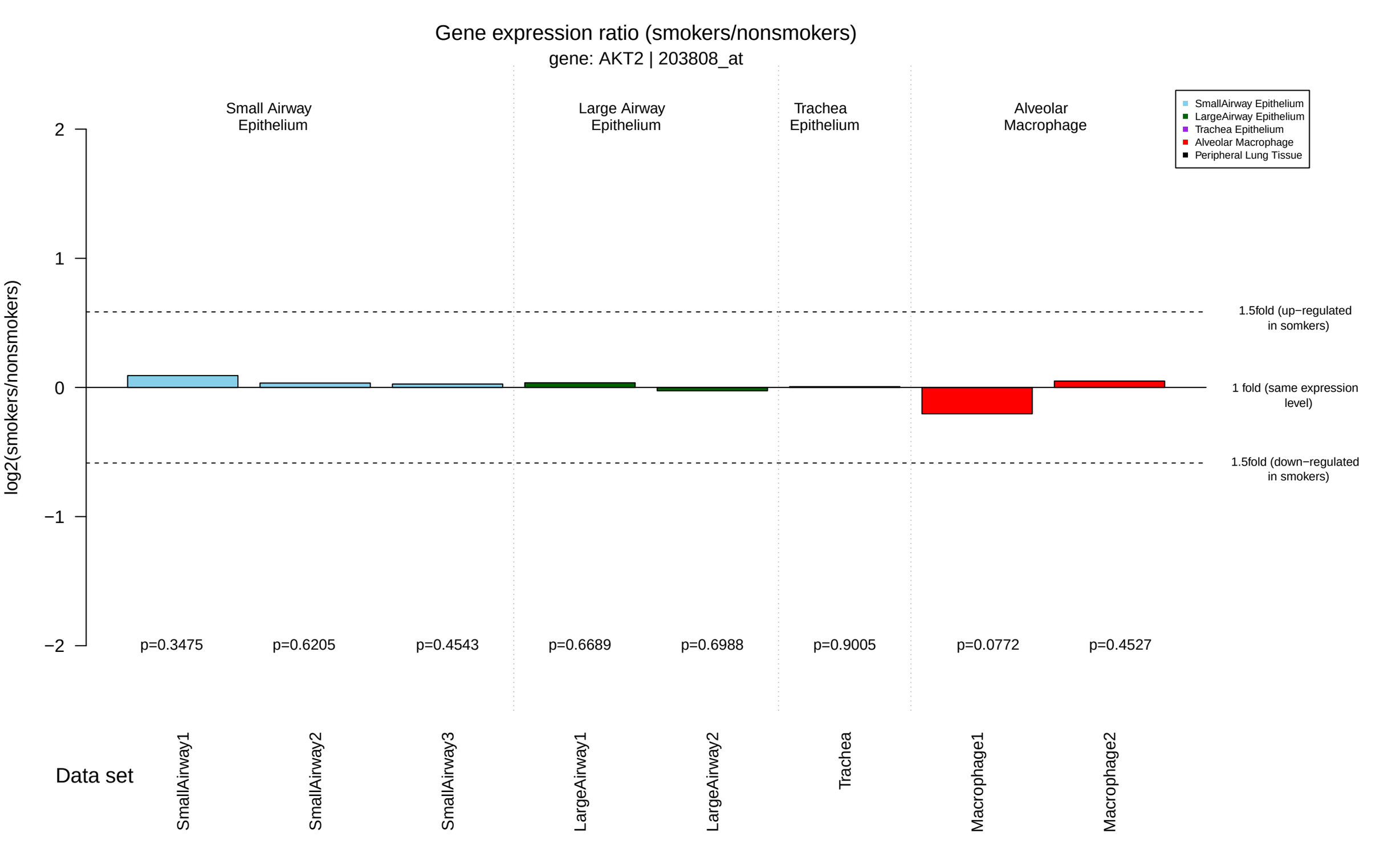
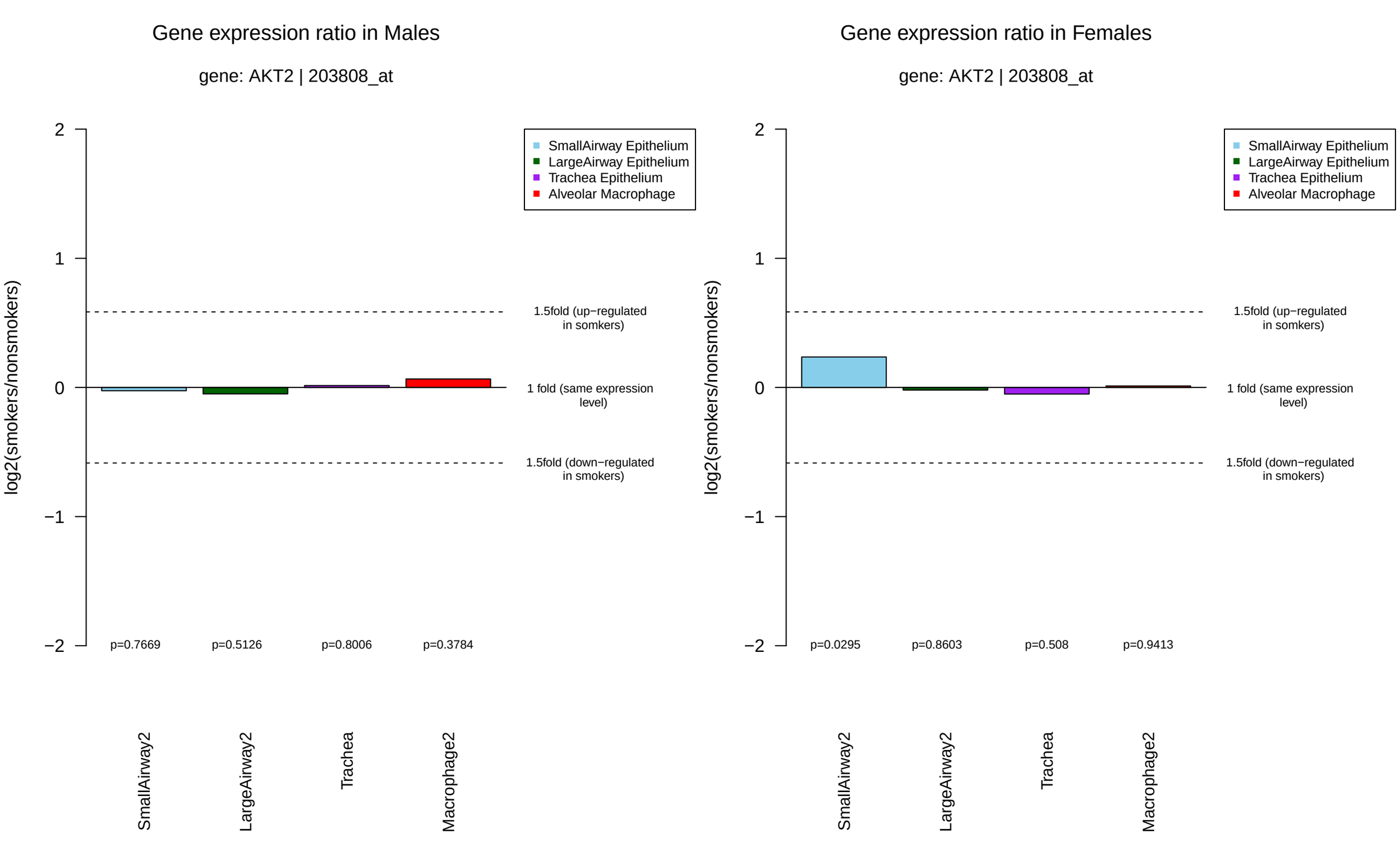
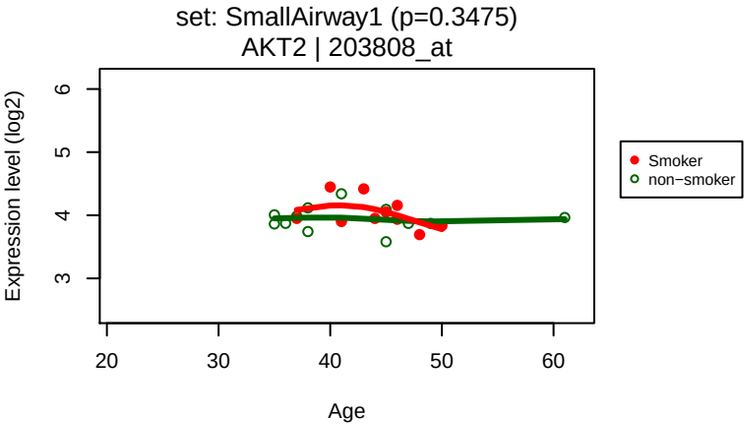
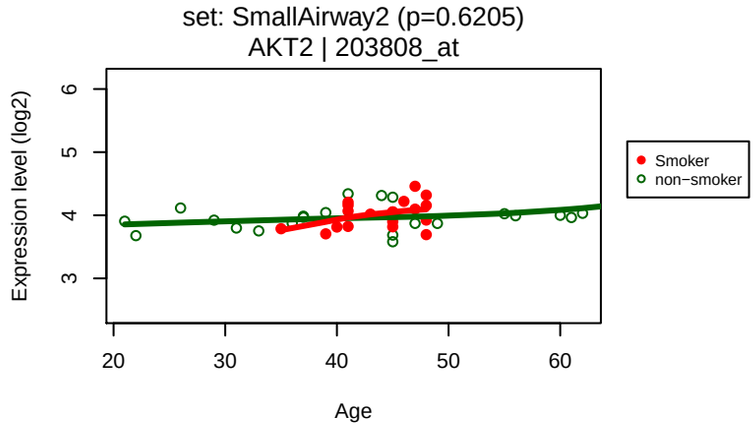
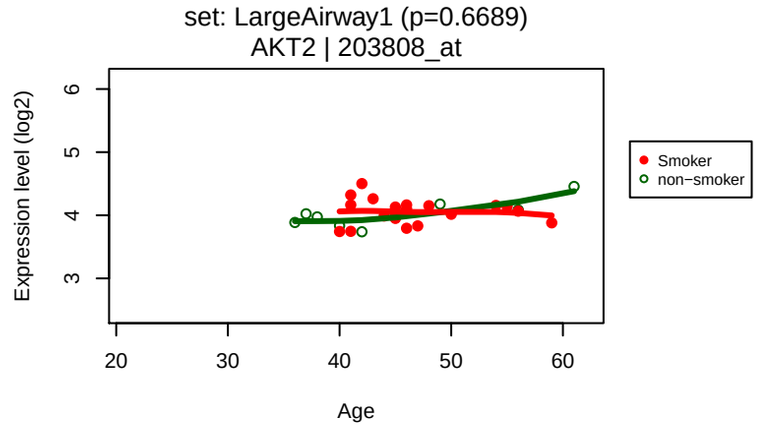

Gene expression of smokers and nonsmokers
gene: AKT2 | 203808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3475
p=0.6205
p=0.4543
p=0.6689
p=0.6988
p=0.9005
p=0.0772
p=0.4527
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

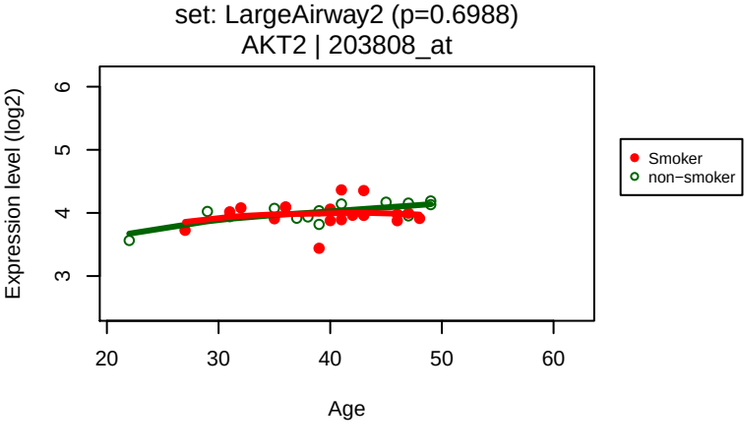

Gene expression ratio (smokers/nonsmokers)
gene: AKT2 | 203808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3475
p=0.6205
p=0.4543
p=0.6689
p=0.6988
p=0.9005
p=0.0772
p=0.4527
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

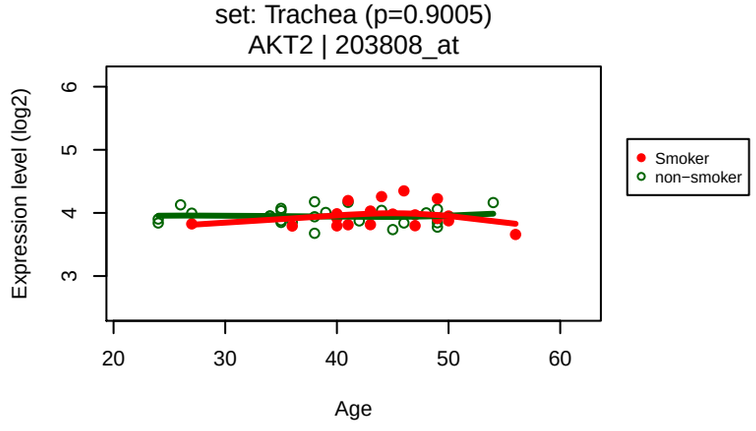

Gene expression ratio in Males
gene: AKT2 | 203808_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7669
p=0.5126
p=0.8006
p=0.3784
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT2 | 203808_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0295
p=0.8603
p=0.508
p=0.9413
SmallAirway2
LargeAirway2
Trachea
Macrophage2

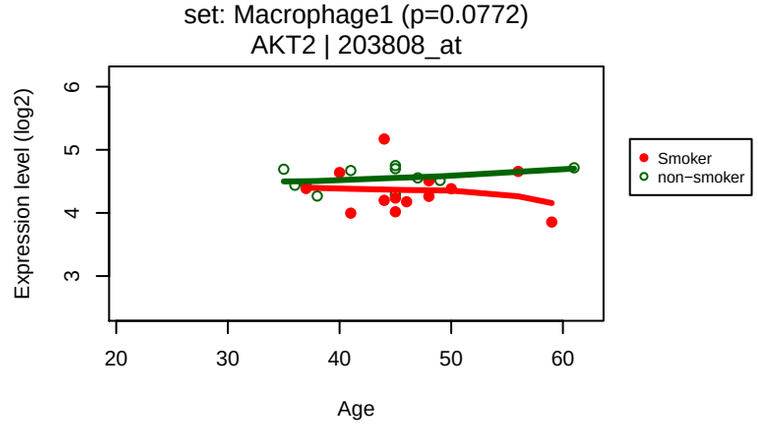

set: SmallAirway1 (p=0.3475)
AKT2 | 203808_at
Expression level (log2)
Age
Smoker
non−smoker

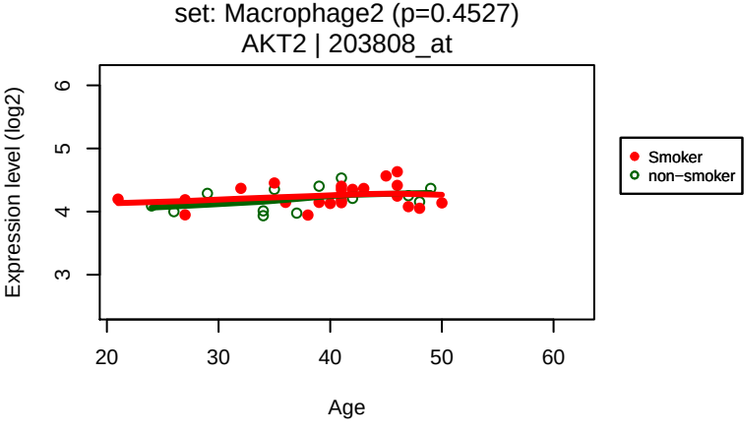

set: SmallAirway2 (p=0.6205)
AKT2 | 203808_at
Expression level (log2)
Age
Smoker
non−smoker

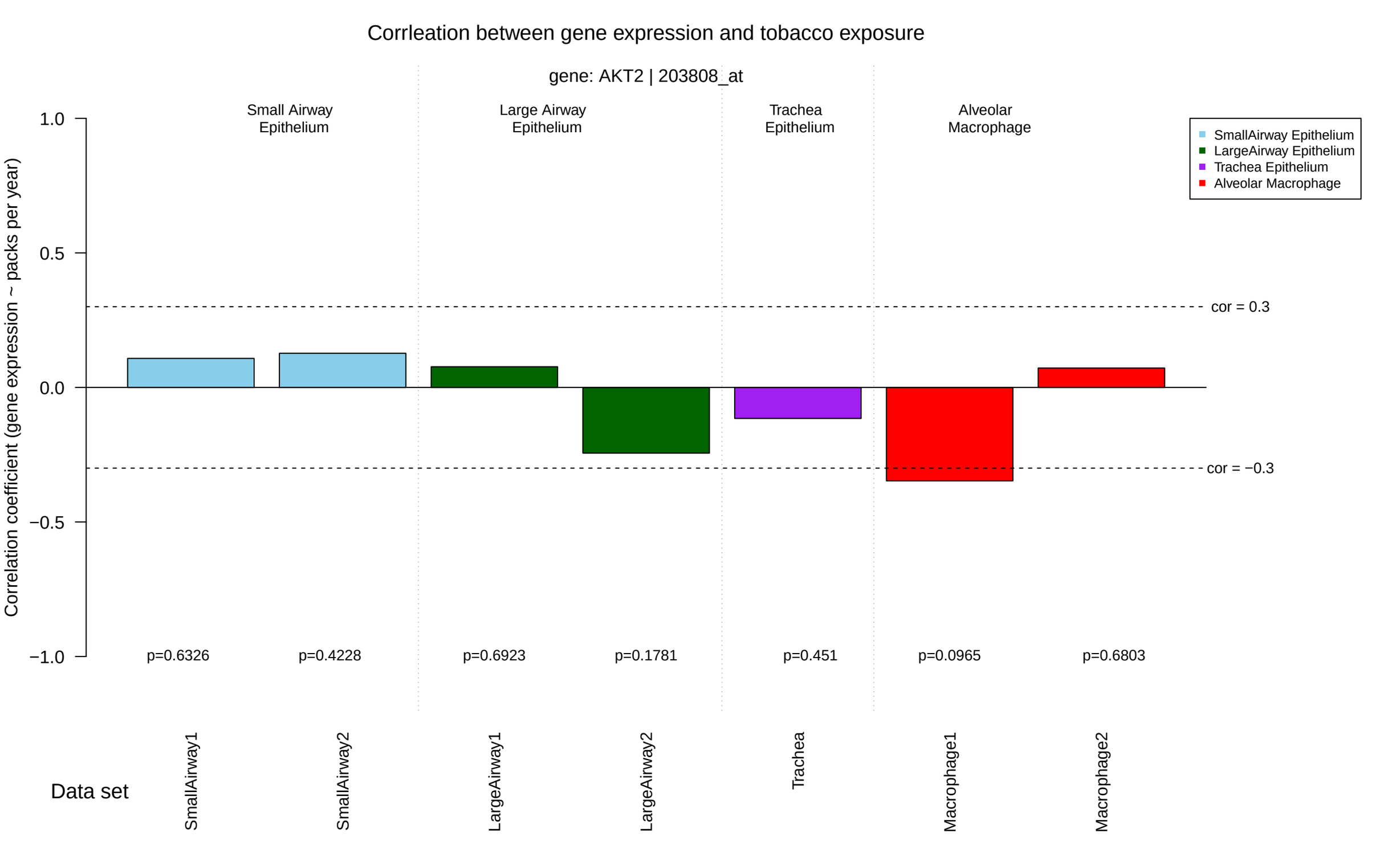

set: LargeAirway1 (p=0.6689)
AKT2 | 203808_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6988)
AKT2 | 203808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9005)
AKT2 | 203808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0772)
AKT2 | 203808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4527)
AKT2 | 203808_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT2 | 203808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6326
p=0.4228
p=0.6923
p=0.1781
p=0.451
p=0.0965
p=0.6803
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2